Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 · UNIT 4 · CLASS 48 Friction Numerical

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 48 Friction Numerical

1 CLASS 48 Friction Numerical

Study focus: CLASS 48 Friction_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000138 · EV0000139

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Problem six slash eighteen The uniform slender bar as an ideal roller at its upper LDA.

Key details

- Determine the minimum value of the angle theta for which equilibrium is possible for mu s point two five and for mu s point five.
- We don't know the length of this bar, we can assume it as L.
- Let capital L be the length of this slender bar.
- And when muus is 0 to 5, then in that case this entire system will be in a state of equilibrium.
- Since it's a roller, there will be only one reaction from the surface onto this roller that will be normal to the surface.
- Next year the slender ball is resting on the surface, so I am going to assume it as a.
- Tough surface there will be two reaction normal reaction N B and one more friction.
- The direction of friction depends upon the tendency of motion.
- Because of this inclination the slender bar will have a tendency to slide this way.
- And we know frictional force will be as equals to mu s into n.
- Let it be L.
- Next, the weight of this center bar will be acting to the geometric center G that will be at a distance of L by 2 with respect to A, L by 2 with respect to B.
- Next, this is W vertically downward.
- W is vertical, F B is horizontal, N B is vertical.
- Whereas N A is inclined.
- For that I should know this angle here.
- Then how to find this angle?

- To find the angle I am going to drop it perpendicular to this point, then extend this horizontal surface here like this.
- Then you can see there is a right angle triangle A.
- C D this right angle travel this is 90 degree this is theta then if you look at this here this angle will be 90 so the remaining angle will be 15 degree if this is 15 degree this angle will be 15 degree.
- This is 15 degree.
- This will also be 15 degree and the component of this na will be right side na cos 15 and the component of NA along y direction will be upward and that will be na sine 15 degree.
- Before that, remember here right now here is point.
- Now the first condition of equilibrium is sigma f of x is equals to 0, considering right side component as positive and left side as negative.
- Start with this NE, NA is towards right side, so plus N A is N A cost 15, F B is left side, so minus F B is equals to 0.
- yes b can be written as 0.25 n b in that case n a will be equals to 0.25 n b1 cos 50 that will be equal to 0.2588 times that of n b I'll call it as equation one.
- Next, second condition of equilibrium sigma f of y is equals to 0 taking upward component positive, downward, negative.
- N component is upward, so plus n mu.
- Sine fifteen degree, then W is downwards so minus W, then N B is upwards so plus N B is equals to C.
- Next we know.
- N is 0.258 times N V from the equation one substitute that here into sine fifteen minus W plus N B is equal to zero.
- Then N B can be written as.

Example or application from the lecture

- This is the figure given.
- After this, let us consider the length of the slender bar.
- Going to consider like this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 05:00–09:59

N V will be equal to W divided by.

Key details

- One plus point two five eight into sine fifteen that will be equal to one point zero six seven zero.
- This will be equation.
- Next to the last condition of equilibrium is moment.
- Here I am going to apply moment about yes 0.
- Next comes the force W.

- W is acting in the clockwise direction, so it is minus W.
- This is force, W is force.
- Into perpendicular distance since the linear fraction of W is vertical, the distance should be horizontal.
- This distance if this is theta, this angle will also be theta, since this is L by two, A G distance is L by two, this will be equal to L by two cos.
- We are done with the moment of this W.
- Next comes F B, F B is acting in the clockwise direction.
- into distance since the line of function of f b is horizontal the distance will be vertical distance from this rectangle triangle A C B this side will become L sin 15 degree that will be the perpendicular distance for this force F.
- N sin 15.
- Next we are left with one more force NB.
- N B is acting in the counterclockwise direction.
- Since the line of action of N B is vertical, the perpendicular distance will be horizontal, that will be this distance.
- This will be equals to N.
- We have N B in the second term, we have N B in the third term also.
- Write the letters as it is L by two cos theta, then minus 12 psi n b sine fifteen degree, then plus of n b cos theta is equal to zero.
- In all the three terms we have N V and L.
- sine theta will be equal to 0.4665 cos theta sine theta by cos theta will be equal to tan theta that will be equal to 1.866 then theta will be equal to 61 point.
- Next question problem six slash twenty three eighty two kg man pulls the forty five kg card up the incline at steady speed.
- Determine the minimum coefficient of static friction for which the man's shoes will not slip.
- Also determine the distance S required for equilibrium of the man's body.
- If you observe this figure, there is an inclined plane.
- The inclination of the plane with the horizontal we can find it out, assume this is theta, then theta will be equals to tan inverse of 3 by 10.
- The inclination of this handle with respect to a line parallel to the incline plane in 40 degree.

Example or application from the lecture

- Moment about which part we have to consider here.
- Once we consider moment about A as 0, considering counterclockwise positive and clockwise negative, moment because of these two will be zero.
- Let's cancel that.

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EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

Core explanation · 10:00–14:59

N I F by N in general.

Key details

- Then what is F?
- F is frictional force and N is normal reaction.
- Then after that we have to find this yes.
- S is the horizontal distance between the A.
- That is the.
- Position of the man's shoe with respect to the center of gravity of man.
- This is also unknown.
- Once we know the force in this handle from the free body diagram of cart, the same force will be acting throughout from this point to the other point.
- normal to the incline plane.
- Next is weight of the cart.
- The self weight of the cart will be always acting vertically downward I'll call it as WC.
- WC stands for weight of the card that is equals to mass of the card 45 into.
- This will be m.
- Next, this is inclined.
- As I said before, this will be theta, so this will also be theta.
- Then theta is equals to tan inverse from the triangle we know 3 by 10.
- Next, after this there will be a force in this hang.
- That force is unknown right now, I will call it as U.
- Then the components of this will be along the positive direction of x axis that will be f cos 40 and one more will be in this way f sin 40 next by considering the free body diagram of cart.
- We want to find only the value of F we only need to find the reaction at these two wheels or rollers.
- This f is in the positive direction of x axis, so it will be plus f cos.
- Then this W Wc the component of this WC is down the incline, so minus WC, since the angle is with respect to y axis, it will become sine.
- As I said before.
- I need to find only the value of f so after simplifying this f will be equal to 165.6 newton.
- It is surface here, it will there will be having two reactions.
- One is normal reaction, one more is frictional force, one is along and one more is normal to the nuclear.
- Then at G the weight of the man will be acting that is vertically downward weight of the man w m will be equals to 82 that is mass of the man into acceleration due to gravity that is 9.81 newton.
- Next there will be a force through this member through the handle that will be equal and opposite to this F.
- But once we come to Wn, this is inclined with respect to y and x axis.
- To resolve this, since this angle is theta, this angle will also be theta this angle will be theta the components are going to be like this will be wm cos theta this is the component along y direction and one more will be w m sine theta.

Example or application from the lecture

- After that, once you consider a man, we don't know the force acting in this handle of the card.
- That's why let's start with the free body diagram of the cart.
- Once we consider a card let's start with x axis that is along the inclined plane and y axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 15:00–19:59

This will be W M sine theta.

Key details

- The component is going to be f cos 40 and one more component will be log y direction f sine 40 d.
- Sigma f of x is equal to zero.
- This n is towards the positive side, so.
- F a is towards the positive side, so plus then f is in the opposite direction.
- then w is w component is also in the opposite direction so minus w sine theta is equals to zero so f then minus f is 165.6 into cos 40 then minus w m is 82 into 9.81 into sine theta is equal to 0.
- Then we know everything except f a so we can calculate f a f a is equal to 358 point nine Newton.
- Next we have to find.
- The one more force that is ne, apply the second condition of equilibrium, sigma f y is equals to 0.
- F into sin 40 is equal to 0.
- Here except NA we know all the other terms, so N A will be equal to 870 6.93 Newton.
- Once we know F A then N A we know mu.
- Coefficient of friction muis will be equal to all this, F A by N A.
- This will be equals to 0.409.
- We got the coefficient of static friction.
- Next we should know the value of yes.
- S we will get only when the moment equation comes because moment is forced into perpendicular.
- Here when we talk about momentum.
- M into perpendicular distances yes.
- here what do we know here is this angle with respect angle here will be theta.
- If this is theta and this is theta 40 this angle will also become theta plus 40 degree.
- Now apply the condition moment about D is equals to zero.
- Since this is in the clockwise direction, it will be minus force is E of sine of.
- Theta plus 14 into perpendicular distances 175.
- Then again it is clockwise so minus of cos.
- theta plus 40 into perpendicular distance will be equal to 850 that is equal to 0.

- Here we know the value of f we know the value of wm we know the value of theta so we just need to find the value of s so s will be equals to 126.2 meters.

Example or application from the lecture

- In the same vein, if this is 40 degrees, it's given here if I draw a line parallel to incline this will be 40 degree.
- So apply the next condition of equilibrium considering moment about any point zero, here we are going to consider moment about this point A as 0.
- we know need to consider these components along X and Y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 20:00–24:59

That's it for today's class.

Example or application from the lecture

- Try to solve the problems given in the cushion bank as well as the assignment.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 48 Friction_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 48 Friction_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 48 Friction_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
CLASS 48 Friction_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00038	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00039	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00040	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00041	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

C Revision and retrieval practice